IRexp and IRSpectra-Bench: redistributable experimental IR band lists, a blind peak-list benchmark, and a recall-bound diagnosis of LLM elucidation

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We release *IRexp* and *IRSpectra-Bench* as chemical-information resources: the largest openly redistributable collection of experimental infrared *band lists* (121,233 records; 43,060 structure-linked; 33,201 full IR + ^{1}H + ^{13}C + structure quadruples), a blind mechanically scored peak-list benchmark of 194 compounds, a fixed RDKit InChIKey-connectivity scoring contract, and decomposable generation-recall / verification-precision metrics others can report. On IRSpectra-Bench, given the molecular formula and peak lists exactly as reported in open-access papers, a frontier large language model recovers the correct constitution for 28% (top-1; 95% CI 22–35), or 15% once reweighted to the corpus composition — far below the near-100% implied by curated demonstrations. The bottleneck is candidate proposal, not verification: the true structure enters the pool for only 34% of compounds, and where it does, training-free forward-verification selects it 89% of the time (58/65; 81% on the 37 compounds where a ranker had a choice). Recovered from published top-*k* figures, the same split carries 68–83% of the accuracy collapse when three systems move from curated or simulated data to real heterogeneous spectra. Generating wider lifts recall 32% $\rightarrow$ 42% and top-1 23% $\rightarrow$ 30% on 60 compounds; verification itself moves whole-benchmark top-1 only 28% $\rightarrow$ 30%. Masking the spectra drops top-1 from 23% to 5%, and Grok 4.6, Gemini 3.7 Flash and GPT-5.6 Sol all verify better than they generate. Frozen predictions, scorers and code are released for mechanical re-evaluation.

1. Introduction

Open experimental infrared data usable for machine learning remain a chemical-information bottleneck. View-only archives such as AIST SDBS[1] hold more structure-linked *spectra* than any redistributable corpus, yet forbid bulk reuse; most public elucidation suites train or evaluate on simulated or single-instrument spectra. Determining structure from spectra is nonetheless central to synthetic and analytical chemistry. Machine learning uses encoder–decoder models on paired corpora; *Spectro*, for one, learns ^{1}H/^{13}C/IR $\rightarrow$ SELFIES from 6,833 molecules[2]. General-purpose LLMs are now reported off-the-shelf: a non-peer-reviewed 2026 industrial white paper found Claude Opus matching or beating commercial NMR-prediction software forward (structure $\rightarrow$ spectrum, $\pm$0.08 ppm ^{1}H) and "recovering all eight simpler structures on every attempt" in the inverse[3]. We treat those numbers as a claim to test, not an established baseline.

That evaluation is narrow: 15 inverse problems on curated single-ring or two-fragment molecules, NMR only, seven given the *starting-material structure* as a hint, and "recovery" scored leniently over three runs and three ranked candidates. The chemist's question — *take an arbitrary experimental spectrum from a paper and recover the structure* — remains open. It needs open experimental data at scale, a blind benchmark with a fixed molecular- representation scoring contract, and honest accounting of which stage of elucidation binds.

We provide all three as reusable chemical-information objects. *IRexp* (§2) is, to our knowledge, the largest openly *redistributable* collection of experimental IR *band lists* (121,233 records; 43,060 structure-linked; 33,201 IR + ^{1}H + ^{13}C + structure); SDBS remains larger for view-only structure-linked spectra but cannot be redistributed (§2.1). *IRSpectra-Bench* (§3) is the blind, mechanically scored peak-list benchmark built from it — 194 compounds, complexity-stratified, multimodal IR + ^{1}H + ^{13}C — with RDKit InChIKey-connectivity scoring and decomposable generation-recall / verification-precision metrics. Concurrent 2026 suites bound related tasks (MolQuest scores 530 post-2025 compounds by exact canonical SMILES[4]; SpecX[5]; NMRGym[6]); ours is an openly redistributable suite on literature-reported peak lists with a stage decomposition measured on Claude, replicated across three other model families (§4.7), with a within-compound solver-methodology control (§4). Ground truth is literature structures resolved deterministically (OPSIN/RDKit) and checked mechanically (560/560 bands on a seed-fixed sample, §2.3; expert-chemist review of elucidation outputs is formally deferred, §8); no LLM curates labels or scores predictions. Priessner *et al.*[7] already note that re-ranking cannot recover a missing candidate; we supply the missing measurement infrastructure at scale.

Forward-verification elucidation (§5) turns the model's *strong* direction (forward prediction) against its *weak* one (inverse regiochemistry); the loop is prior art (§1.1), the decomposition ours, run training-free on every compound. The finding is sharp asymmetry: given a candidate set containing it, the model *verifies* the correct structure 89% of the time yet *proposes* it for only 34%, so generation binds the result (Fig. 1). Gain is bounded (top-1 28% $\rightarrow$ 30% on n=194; 23% $\rightarrow$ 30% on the 60-compound arm where wide generation was also tested) and, by our own measurements, cannot exceed a recall/precision ceiling without sharper verification or 2D-NMR data.

Contribution. We release IRexp and IRSpectra-Bench — redistributable experimental band lists, a fixed blind peak-list protocol, and decomposable recall/verification metrics others can report on a shared InChIKey scorer — measured on every compound (34% proposed; 89% selected once proposed), with a four-vendor replication of that split. The paper fills an infrastructure and accounting gap in cheminformatics: it diagnoses where elucidation binds; it does not claim a solved elucidator.

Solver and verifier (Fig. S1) are LLM agents on a consumer subscription, with no fine-tuning and no API spend. Inference is not exactly reproducible — no pinned snapshot, temperature or seed (§9) — but scoring is: predictions, ground truth and scorers are released, and every training-free number regenerates from them. The HOSE lookup, GNN (§5.4) and trained generator (§5.6) are fenced complements.

1.1 Related work

Four camps surround this work; IRexp/IRSpectra-Bench occupy a distinct wedge — *redistributable experimental IR band lists*, a *blind literature peak-list protocol*, and *stage-decomposed metrics* — rather than a claim to state-of-the-art agent accuracy.

Trained spectra → structure models. Sequence and graph decoders on paired corpora — *Spectro*[2], multitask 1D-NMR models[8], NMRTrans/NMRSpec[9], *NMIRacle* (IR + ^{1}H + ^{13}C)[10], Alberts IR transformers[11,12] — are accurate in-distribution but retrained per modality and typically scored end-to-end. IRexp (§2.3) complements NMRSpec with redistributable IR band lists; neither corpus alone is multimodal. Upstream IR work from our group (*vIR-OLO*[13], *j-IR-vis*[14]) identifies functional groups but does not generate candidate structures. We do not score Spectro, NMIRacle or CASE on IRSpectra-Bench (§8); the resource is designed so those baselines can be added under one scorer.

LLM agents and puzzle benches. LLM agents already plan syntheses (ChemCrow[15], Coscientist[16]) and call elucidation routines; we measure what one returns. Off-the-shelf and multimodal LLMs[3,17–19], puzzle benchmarks (MolPuzzle[20]), and agentic IR interpreters (IR-Agent[21]; Priessner *et al.*[7]) improve *how* a model reads spectra or re-ranks a fixed pool. Cross-paper numbers swing with harness: GPT-4o scores 1.4% on MolPuzzle[20], 27.8% with chain-of-thought, 57.8% with tree-search[19]. We fix one pre-registered RDKit-InChIKey protocol with bootstrap CIs and ask which stage binds once the spectra have been read.

Contamination-aware and heterogeneous-data benches. Most prior suites use simulated or single-instrument spectra[2,10–12]. IRSpectra-Bench uses literature-reported peak lists across thousands of laboratories and reports formula-only and recency controls (§4.6). Espejo Morales *et al.* frame agentic search on raw instrument files: 80.9% on education spectra, 20.6% on 34 industrial samples[22] — they ask which architecture; we ask which stage binds on published reports (§8).

Forward verification, CASE and prior loops. Matching predicted to observed shifts — DP4[23,24], NMR crystallography[25,26], CASE[27,28] — is easier than inverse generation. CASE enumerators have exhaustive recall by construction; the LLM literature almost never reports whether the true structure was proposed at all (§6). *NMR-Solver*[29] runs the same generate-and-verify loop without an LLM (52.9% top-1 on literature ^{1}H/^{13}C, formula supplied; 31.6% with stereochemistry). We decompose error rather than beat that score. *NMRAgent*[30] is a complementary best-effort system. Kliuev *et al.* place the best of six LLMs at 24% on 105 peak lists[31]. Priessner *et al.* already state that re-ranking cannot recover a missing candidate[7]. We add measurement at scale: the split on 194 compounds across four model families and four verifiers, recovered from published top-*k* figures (§6).

2. The IRexp dataset

2.1 Motivation

An IRexp record is a *band list* — peak positions in cm^{-1} transcribed by an author into a paper's text — with co-reported ^{1}H/^{13}C shift lists and, where resolvable, the 2D structure. IRexp contains no absorbance traces. That is the published form a language model consumes, but a different object from a digitised spectrum, and the two should not be counted against one another.

Among *digitised spectra*, free downloads run to the NIST WebBook[32] (≈16k) and the Chemotion ELN deposit[33] (≈2k). AIST SDBS[1] is larger (≈54k FT-IR, all structure-linked) but *view-only* (50 spectra/day, no bulk export), so IRexp does not compete there: SDBS alone holds more structure-linked spectra than IRexp holds structure-linked band lists. Among text-derived *band lists* IRexp is the largest *openly redistributable* collection by record count — deliberately scoped to that object type.

2.2 Construction

Experimental sections report per-compound IR band lists with co-reported ^{1}H/^{13}C NMR in a stable textual convention. Mining it is mature[34]; we add a pipeline specialised to the *IR band list*, the modality those tools passed over.

- **Discovery.** Open-access literature is harvested in bulk from the PMC Open-Access Subset[35] and the Chemotion FT-IR deposit (RADAR4Chem).
- **Extraction.** A deterministic parser segments experimental text per compound and extracts IR wavenumbers and ^{1}H/^{13}C shifts, gated against scan-range artefacts and prose false-positives.
- **Resolution.** IUPAC names go to SMILES with OPSIN[36] and RDKit[37] canonicalisation (InChIKey, SELFIES[38]), with a PubChem[39] fallback for trivial names; handling open-access label conventions and narrative artefacts raised structure coverage from 24% to 35%.

2.3 Contents and licensing

Table 1 summarises contents and provenance.

The pools differ: PMC records are author-transcribed (median 9 bands), Chemotion records peak-picked from deposited spectra (median 39); users training on band density should treat them apart. `scripts/split_license_pools.py` reports provenance (`source_doi`) and stamped `license_pool` splits. **Licensing.** A Europe PMC join over all 15,416 unique IRexp PMCIDs stamps every record (`scripts/join_pmc_licences.py`): **87,617** commercial (CC-BY/CC0), **20,938** CC-BY-NC*, **1,897** ShareAlike (Chemotion + rare PMC SA), and **10,781** empty/unknown (excluded from the commercial Zenodo pool). Chemotion records are CC-BY-SA-4.0. Do not treat the undivided PMC provenance slice as a single CC-BY-4.0 redistribution — use `license_pool == "commercial"` (or `data/irexp/licence_pools/`).

Fig. 1. Diagnosis on IRSpectra-Bench (n=194): generation recall, not verification, is the bottleneck. Where the true structure is never proposed, no ranker can recover it; where it is proposed, verification usually selects it. End-to-end top-1 is the product of those two rates, which have different denominators and are not differenced.

Table 1. IRexp dataset contents and provenance.

field	value
experimental IR band-list records	**121,233**
…co-reporting ^{1}H and/or ^{13}C NMR	87,075 (72%)
…with a resolved 2D structure	43,060 (35.5%)
…of these, with ^{1}H and/or ^{13}C NMR	40,702
…of these, with both ^{1}H and ^{13}C (full quadruples)	33,201
by provenance: author-transcribed from PMC-OA text (mixed CC terms; see Licensing)	119,345
by provenance: peak-picked from Chemotion ELN deposits (CC-BY-SA)	1,888

Extraction fidelity is measured. On a seed-fixed random sample of 60 PMC-sourced records (`scripts/audit_extraction.py --n 60 --seed 0`) we re-fetched each article and checked every recorded wavenumber against its text: 560/560 bands and 60/60 records were confirmed (Wilson 95% CI 99.3–100% and 94–100%). This bounds *transcription* error — hallucinated, mis-parsed or unit-mangled values — below 1% of bands. It does not measure whether the parser found every IR string in every paper — a recall question requiring human reading; that extraction-recall audit is formally deferred (§8).

`irexp_resolved` (43,060 records, 100% structure-linked) is the benchmark-ready split, ≈6× the 6,833-molecule set used to train Spectro[2] (Fig. S2).

Reuse. Each record carries `source_doi`, stamped `license / license_pool`, and, where resolved, molecular representations (canonical SMILES, InChIKey, SELFIES). Fine-tuning against IRSpectra-Bench should use `data/train_no_bench.jsonl.gz` (or rebuild with `contrib/generator_probe/build_exp_manifest.py`) so benchmark InChIKeys are withheld. Commercial redistribution should use the commercial pool; Chemotion/SA rows remain CC-BY-SA-4.0. Cite the Zenodo deposit once minted and attribute sources via each record's `source_doi` (see Data availability; Hugging Face mirror `ilkhamfy/IRexp`).

3. Benchmark design (IRSpectra-Bench)

From `irexp_resolved` we draw *IRSpectra-Bench*, 194 blind elucidation problems, each giving the *molecular formula* (as from HRMS), the *IR band list* and the *^{1}H and ^{13}C shift lists*; no name, SMILES or scaffold hint. In 10 of the 194 the authors' peak assignments name a ring system (`2CH2·pyrrolidine`, `pyridazinone H5`). Those compounds are solved 1/10 against 54/184 (29.3%) for the rest (`scripts/prompt_leakage.py`), so the annotation does not carry the headline; we keep them rather than drop them. Main-round ground truths are spectrally validated by an automated RDKit check (^{13}C peaks vs symmetry-unique carbons, formula match, SELFIES round-trip) excluding merged or incomplete spectra — 6/140, leaving 134; `scripts/validate_benchmark.py` applies the same filter to all three rounds, regenerating every `clean_qids.json` from the released questions and answers.

The filter does not gate on ^{1}H. In 13 of the 194 retained records the reported ^{1}H integral exceeds the reference structure's hydrogen count — residual solvent, water, exchangeable protons, or a rotamer mixture — printed as a diagnostic, not excluded; the cohort was fixed in advance, and re-filtering post hoc on a criterion chosen after seeing results is a degree of freedom a benchmark should not take. We report the sensitivity instead: dropping all 13 moves the headline from 28.4% to 29.3% (53/181), +0.9 points, leaving every conclusion unchanged. The controlled rounds' 60 compounds are fixed, preregistered sets, audited the same way but reported rather than filtered (57/60 pass; §8).

Difficulty is declared in advance, as a property of the compound. By RDKit ring analysis a compound is *simple* iff it has at most two rings, no fused/spiro/bridgehead system and ≤22 heavy atoms; all others are *complex* (98 simple / 96 complex). The threshold is not load-bearing: sweeping it from 18 to 26 leaves the simple-minus-complex top-1 gap inside a 36–40-point band throughout (36.1 at the narrowest, 39.6 as released; `scripts/difficulty_sensitivity.py`). This axis is distinct from the continuous size gradient of §4.1 (≤15 / 16–25 / >25 heavy atoms), and InChIKey de-duplication across rounds prevents leakage.

The 50/50 balance is by design. The sampler fills each difficulty stratum to half the round; the eligible corpus is 17.5% simple / 82.5% complex. Reweighted to that composition, top-1 falls 28.4% → 15.2% [10.6–20.4] and recall 33.5% → 19.8% [14.4–25.8]; verification precision must be reweighted too (89.2% → 71.5% [50.2–92.5]). We report the benchmark figure as released and the reweighted figure as the better estimate on an arbitrary paper.

Scoring is mechanical (the leaderboard contract). A prediction is *correct* if its RDKit InChIKey connectivity layer (first 14 characters) matches the reference: we score *constitution*, so correct constitution with wrong stereochemistry counts as correct. The *full*, stereochemistry-sensitive InChIKey gives 21.1% top-1 and 25.8% recovered (41/194 and 50/194) against 28.4% / 33.5% — 7.3 points lower on top-1 and 7.7 on recovered — so the constitution figure is an upper bound on full-stereochemistry accuracy. Constitution is the headline because 1D ^{1}H/^{13}C/IR rarely fixes absolute configuration: only 10.3% (20/194) of answers carry a *defined* (assigned R/S) stereocentre, and a model is penalised there for information the prompt never contained (`scripts/score_main.py --stereo`). External submissions follow the same contract via `scripts/score_submission.py` and the protocol in `docs/LEADERBOARD.md`: up to three ranked SMILES per `qid`, no structure hints, document model version and tool access.

Metrics. *Top-1 (exact constitution)*: best-ranked candidate matches at the connectivity layer; *recovered (top-3)*: reference among up to three returned candidates, the lenient "recovery" protocol of ref.[3]. *Generation recall*: reference in the candidate pool *before* re-ranking, the ceiling any verifier can reach (coincides with recovered except in §5.3, where the pool is larger). *Verification precision (conditional on recall)* over recall-positive compounds isolates verification from generation. The forward-verifier ranks by symmetric *chamfer distance* between predicted and observed ^{13}C peak sets (lower is better); Morgan(2, 2048) Tanimoto[40] gives a graded scaffold-family signal. CIs are bootstrap 95%; model-vs-model differences use McNemar's exact test[41] with Holm correction[42]. Reports on IRSpectra-Bench should quote all three primary numbers — top-1, generation recall, and verification precision | recall — not top-1 alone.

Solvers are frontier LLM agents (Claude Opus), one subagent per batch, closed-book under a consumer subscription: no tools beyond an RDKit formula check, no ground truth, verified by grep-auditing transcripts. Formula-adherence rates by round are reported in the Electronic Supplementary Information.

4. How well do LLMs elucidate real structures?

Having defined IRexp and IRSpectra-Bench as reusable data objects, we next report the decomposable metrics they enable — top-1 constitution, generation recall, and verification precision — rather than a single end-to-end accuracy claim.

4.1 Headline performance

Solver agents work blind from formula + IR + ^{1}H + ^{13}C, returning up to three ranked candidates in bounded contexts reset every 2–12 compounds rather than one long context — the variable §4.3 shows what matters. The benchmark comprises all 194 compounds (134 spectrally-validated + 60 controlled-round; `data/benchmark_main/raw/`).

A strictly-validated subset (134 main-clean + 57 controlled-clean = 191) reproduces this within ≈1 point on every metric — top-1 28.8%, 95% CI 23–36; recall 34.0%; simple 49.0% / complex 8.4% — so the asymmetric inclusion of pre-registered controlled sets does not drive it. The deliberate 50/50 difficulty balance does: reweighted to the 17.5%-simple composition of the eligible corpus, top-1 is 15.2% [10.6–20.4] and recall 19.8% [14.4–25.8] (§3). Per-stratum figures are unchanged; the reweighted number is what to read as performance on an arbitrary paper. Accuracy falls monotonically with size (Fig. 2): 60.5% top-1 at ≤15 heavy atoms, 28.3% at 16–25, 7.0% above 25 (Fig. S3).

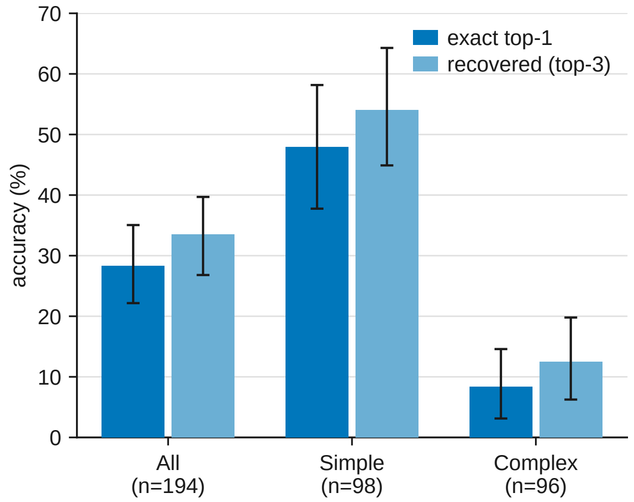

Fig. 2. Top-1 and recovered accuracy on IRSpectra-Bench by difficulty (all / simple / complex, n=194), with bootstrap 95% confidence intervals. The benchmark separates a realistic difficulty range: simple targets are solved four to six times as often as complex ones on both metrics.

Of the 137 analysable top-1 misses (139 in all; two predictions did not parse; `scripts/analyze_misses.py`), 76.6% are constitutional isomers of the true structure against 23.4% with the wrong formula — not mostly regiochemistry: only 22.6% share the true Murcko scaffold, 2.9% reach Tanimoto ≥0.85, and the median isomeric-miss Tanimoto is 0.39. Strict regiochemistry accounts for a fifth of failures; most misses are larger rearrangements of the same atoms. Near-degenerate ^{1}H/^{13}C shifts under-determine connectivity; the misses show *wrong connectivity at the right composition*.

4.2 Reconciling with prior reports

Our 28% top-1 sits far below curated NMR-only demos[3] and below in-distribution trained baselines (48–94%)[2,10,12]: difficulty, scoring leniency, starting-material hints and hand curation inflate those reports, while settings differ on every axis from our blind literature spectra. The ≈40× MolPuzzle swing for one model (§1.1) bounds what unaudited near-100% claims can bear.

4.3 Methodology dominates: a within-compound control

On the same 20 molecules, recovered rose 5% → 15% (and top-1 0% → 15%; McNemar p=0.25) when four bounded, tool-using agents replaced one long context — directional only.

4.4 Model comparison: the benchmark orders capability but separates only the extremes

We solved a fixed 24-compound subset blind with four Claude models spanning a wide capability range, including the newest (Fable 5), under one prompt, one scorer and one candidate budget (Fig. 3, Table 3).

One protocol asymmetry must be disclosed: the three comparison models solved the 24 compounds as four six-compound contexts each, whereas the Opus column reuses

Table 2. Headline elucidation performance on IRSpectra-Bench (n=194). Overall and by difficulty stratum, with bootstrap 95% CIs.

metric	overall (n=194)	simple (n=98)	complex (n=96)
top-1 exact constitution	**28.4%** [22–35]	48.0% [39–57]	8.3% [3–15]
recovered (top-3)	33.5% [27–40]	54.1% [44–63]	12.5% [6–20]
scaffold-level (best Tanimoto $\geq$0.45)	56%	73%	39%
mean best Tanimoto	0.59	0.73	0.45

the headline run, where those items sat in one six-compound and two twelve-compound contexts — the variable that §4.3 measures at 5% $\rightarrow$ 15%. The Opus estimate is therefore not protocol-matched and, if anything, handicapped by the longer contexts; this touches neither the nesting nor the Fable-vs-Haiku contrast, and a clean re-run is outstanding in `docs/MODELS.md`.

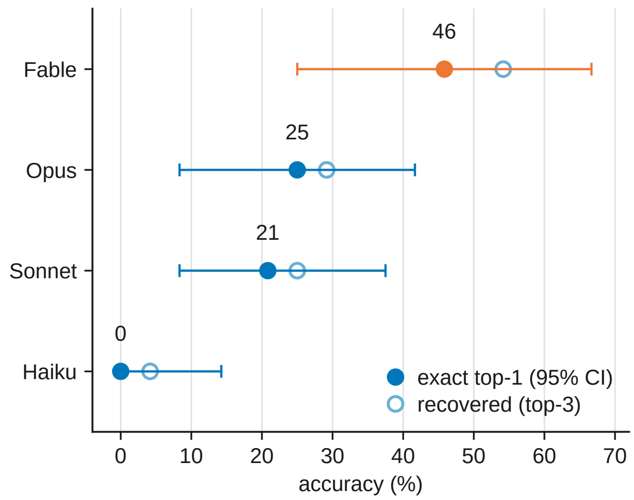

Fig. 3. Four-model comparison on a fixed 24-compound subset, solved blind under one protocol. Outcomes are strictly nested — each stronger model solves a superset of the weaker one's compounds — so the benchmark is capability-sensitive, but at n=24 it is underpowered to separate adjacent models (§4.4).

CIs are bootstrap except Haiku's: Clopper–Pearson exact for 0/24, where the percentile bootstrap is degenerate.

Outcomes are strictly nested (Haiku $\subset$ Sonnet $\subset$ Opus $\subset$ Fable); only Fable-vs-Haiku reaches significance (Holm p=0.006). Opus headline runs used longer contexts than the comparison models (§4.3).

4.5 Domain case study: battery-electrolyte chemistry

IRSpectra-Bench-Electrolyte (46 scored compounds across six electrolyte functional classes, held out and solved blind) matches the headline regime: top-1 26%, recovered 28%; the true structure enters the pool for 13/46 and ranks first in 12/13 — recall-bound, not domain-specific (Fig. S4, Table 4). Per-class intervals overlap (χ^2 p$\approx$0.56); sp^3-C–F reaches 50%, sulfonyl and nitrile 12%.

Intervals overlap throughout (six-class χ^2 p$\approx$0.56; best-vs-worst Fisher exact p$\approx$0.28), so the ordering is hypothesis-generating, not established — C–F couplings pin sp^3-C–F regiochemistry, while sulfonyl and nitrile targets turn on oxidation-state ambiguity and heteroaromatic substitution that $^1H/^{13}C$ shifts underdetermine. It is the recall-bound failure of §4.1 in a domain-specific guise, and it is where §5's *forward-verification* recipe plays a role *analogous* to (not a replacement for) computational-NMR validation.

4.6 Is the model reading the spectra? A formula-only control

We reran the blind protocol with spectra masked, leaving only the formula ([43]).

Outcomes are perfectly nested (McNemar p=0.001). Accuracy is flat in source-paper year across all 194 (r=—0.007; panel b of Fig. 4), bounding pretraining recall. We claim a strong bound, not exclusion (§8).

4.7 Does the diagnosis hold outside one vendor? A four-vendor replication

Grok 4.6, Gemini 3.7 Flash and GPT-5.6 Sol solved the same 60-compound arm under the identical protocol (Table 6). Verification precision exceeds generation recall in every arm (panels c–d of Fig. 4); bootstrapping separates the paired gap for Claude (+52.5 points), GPT-5.6 Sol (+26.3) and Gemini (+23.3); Grok's gap (+9.2) is directional. Three models beat Claude on recall; candidate budgets differ (ours 2.20 vs 3.00 per compound), so recall rankings are approximate. A clean-clone control for Grok (recall 28/60 vs 32/60, McNemar p=0.39) bounds key leakage. Models ran through a coding-assistant harness with undisclosed reasoning tiers (§8).

5. Forward-verification elucidation

5.1 Method

The inverse direction is the model's hard, isomer-blind direction; the forward direction (structure $\rightarrow$ spectrum) is its easy, accurate one[3]. We close a training-free generator–verifier loop:

> *generate* candidate structures (inverse) $\rightarrow$ *forward-predict* each candidate's ^{13}C spectrum, blind to the observed spectrum $\rightarrow$ *re-rank* candidates by the distance between predicted and observed ^{13}C $\rightarrow$ return the best match.

The inverse solver's 373 deduplicated candidates across 194 targets are forward-predicted in shuffled, anonymised batches, blind to the observed spectrum and target identity. Predicted and observed ^{13}C peak sets are compared by symmetric chamfer distance (Fig. 5).

Regioisomers separate by a median 1.21 ppm in forward prediction, but 82% lie within the predictor's $\approx$2 ppm error — a thin margin confirmed against a derangement null at p=0.001 (§5.5). Sharper predictors (GNN, DFT-coupled models[[44];[45,46]]) would move the verification ceiling, not search.

5.2 Result

The first column below is the 60-compound arm (v3 + v2-control) on which §5.3–§5.5 build; the second, the full benchmark. All 373 candidates were forward-predicted — nothing

Table 3. Four-model comparison on a fixed 24-compound subset. All four models ran the identical blind protocol.

model	top-1	recovered	top-1 95% CI
Claude Fable 5	**46%**	54%	[25–67]
Claude Opus	25%	29%	[8–42]
Claude Sonnet	21%	25%	[8–38]
Claude Haiku	0%	4%	[0–14]

Table 4. Per-class performance on IRSpectra-Bench-Electrolyte (n=46).

electrolyte class	n	top-1	95% CI	recovered (top-3)	95% CI
sp^3-C–F	8	**50%**	[22–78]	50%	[22–78]
carbonate	7	29%	[8–64]	43%	[16–75]
phosphoryl	8	25%	[7–59]	25%	[7–59]
glyme / oligoether	7	29%	[8–64]	29%	[8–64]
sulfonyl / sulfonate	8	12%	[2–47]	12%	[2–47]
nitrile	8	12%	[2–47]	12%	[2–47]

Table 5. Formula-only control. Paired on the same 60 compounds as §5.

condition	top-1	recovered (top-3)
formula only	**3/60 (5%)**	3/60 (5%)
formula + IR + ^{1}H + ^{13}C	14/60 (23%)	19/60 (32%)

Table 6. Cross-vendor decomposition on the 60-compound arm. Recall and precision have different denominators, so the criterion is the inequality rather than a difference.

model	generation recall	verification precision | recall	multi-candidate only
Claude Opus (Table 7)	19/60 = 32% [21–44]	16/19 = 84% [62–94]	10/13 = 77%
Grok 4.6	32/60 = 53% [41–65]	20/32 = 62% [45–77]	20/32 = 62%
Gemini 3.7 Flash	30/60 = 50% [38–62]	22/30 = 73% [56–86]	22/30 = 73%
GPT-5.6 Sol	25/60 = 42% [30–54]	17/25 = 68% [48–83]	16/24 = 67%

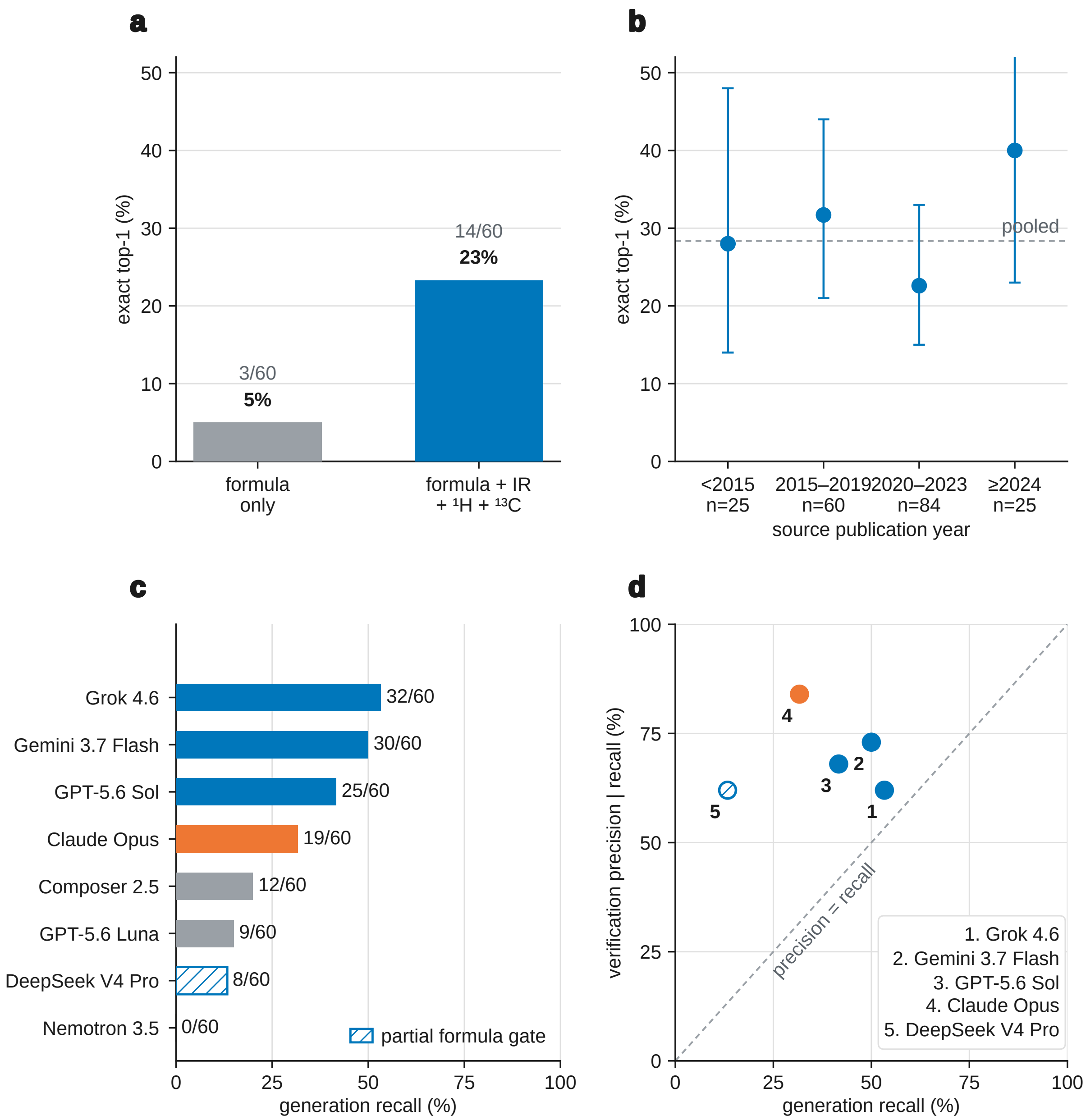

Fig. 4. Robustness of the recall-bound diagnosis. (**a**) Formula-only vs full modality on 60 compounds. (**b**) Accuracy vs source-paper year (n=194; point-biserial r=—0.007; dashed line = pooled 28%). (**c**) Generation recall by model on the 60-compound arm (orange = Claude Opus headline; grey = below formula-adherence gate). (**d**) Verification precision vs generation recall (numbered key; §4.7).

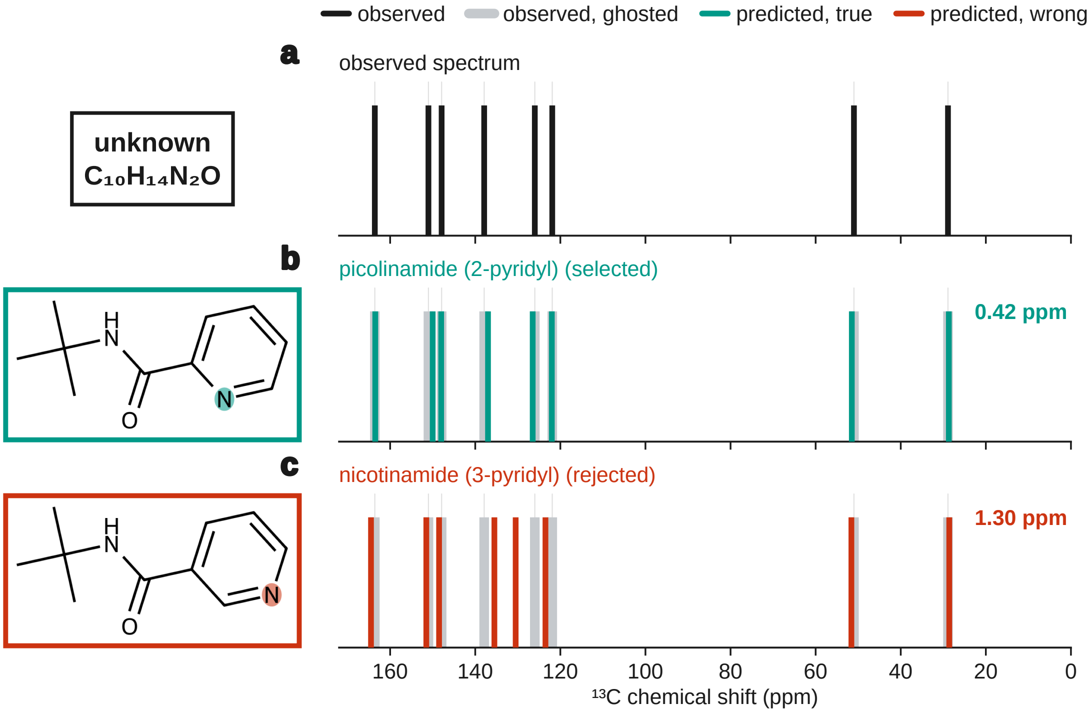

Fig. 5. Forward-verification on a benchmark regioisomer pair: picolinamide and nicotinamide are indistinguishable to the inverse task, but forward-predicted ^{13}C sticks separate the true isomer from the alternative (§5.1).

here is a lower bound; the self-ranking row re-derives Table 2 from §4's output.

When the true structure is among the candidates, forward-verification selects it in 58 of 65 cases (89%). Of the 65, 28 had a single candidate — scored by construction by any ranker — so the pooled figure is not the verifier's margin over chance. Where a choice existed (n=37), forward-verification gets 30/37 (81%) against a 54.0% derangement floor (§5.5) — +27.1 points; self-ranking 27/37 (73%).

The margin over self-ranking remains small and unresolved: seven compounds gained, four lost, McNemar exact p=0.55. We do not claim it. Precision is high in absolute terms while recall binds: top-1 moves only 28% → 30% because the true structure was never proposed for 129 of 194 compounds, which no re-ranking can repair. The verifier converts recall almost completely on *simple* targets (50/53, 94%) and ties self-ranking on *complex* ones (8/12, 67%). Elucidation factorises into two near-independent levers: a strong verifier and a generator at 34% recall that is the wall (Fig. 1).

5.3 Generate-wide: testing the recipe

The decomposition implies *generate wide, verify by forward prediction* — a chemistry analog of self-consistency sampling[47]. Ten solver agents proposed up to six regiochemistry-aware candidates per compound, pooled with the originals and re-ranked.

Wide generation lifts recall 32% → 42% and top-1 23% → 30% (McNemar p=0.34; Fig. 6). Recall plateaus at 42% on polycyclic targets; verification precision falls 84% → 72% — the training-free ceiling. Roughly a third of misses need only regiochemistry around a correct scaffold; two thirds need a scaffold never proposed.

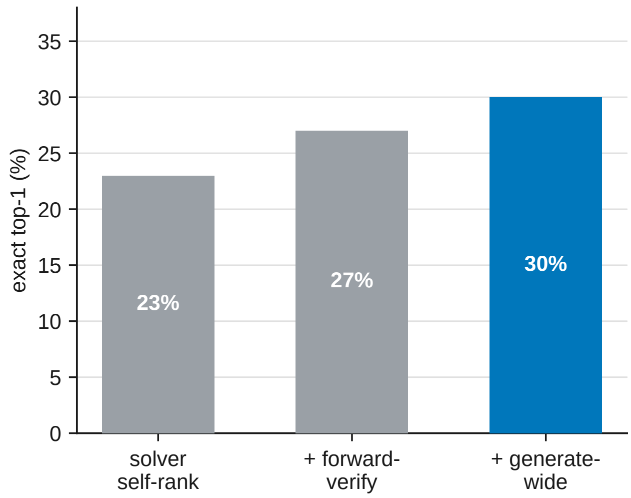

Fig. 6. Forward-verification inference ladder on the 60-compound arm (§5.3). Each stage adds a check on the same compounds; the hero bar is generate-wide top-1.

Table 7. Forward-verification decomposition. Original arm and full benchmark.

	60-compound arm	full benchmark (n=194)
generation recall	19/60 (32%)	65/194 (34%)
top-1, solver self-ranking	14/60 (23%)	55/194 (28%)
top-1, forward-verified re-ranking	16/60 (27%)	58/194 (30%)
conditional on recall — self-ranking	14/19 (74%)	55/65 (85%)
conditional on recall — forward-verification	16/19 (84%)	58/65 (**89%**)
…single-candidate compounds within that set	6/19	28/65
multi-candidate only — self-ranking	8/13 (62%)	27/37 (73%)
multi-candidate only — forward-verification	10/13 (77%)	30/37 (81%)

Table 8. Generate-wide vs original. On the 60-compound arm only, since wide generation was run there; the "original" column is Table 7's first.

	original (60-compound arm)	generate-wide
generation recall (true structure among candidates)	32%	**42%**
forward-verified top-1	27%	30%
verification precision (conditional on recall)	84%	72%

5.4 Non-LLM verifiers: a deterministic lookup and a learned model

§5.3 suggests replacing the verifier with a non-LLM ^{13}C predictor. We tested a HOSE-code[48]-style lookup and a small message-passing GNN, both trained on the same nmrshiftdb2 dump[49] and applied to the same §5.2 candidate sets so only the predictor changes. The GNN is deliberately modest; sharper purpose-built ^{13}C models exist[44,45,50].

The HOSE lookup ties self-ranking; the GNN reaches 59/65 (91%) vs the LLM's 58/65 (89%), directionally (Table 9, Fig. S6). Wagen's MagNET-in-the-loop study[51–53] moves connectivity 46.4% → 55.4% on eight problems — same order as our ladder, underpowered at n=8.

5.5 Negative control

Deranging observed ^{13}C pairings collapses verification precision 89.2% → 73.8% mean (one-sided p=0.001); on multi-candidate sets alone, 81.1% vs a 54.0% floor (+27.1 points). Chamfer margin does not support selective prediction (flat top-1 across coverage fractions) — a reported null.

5.6 Is the recall wall task-intrinsic? A trained-generator probe

That ceiling is a property of training-free LLM elicitation, not the task. Pooling a small IRexp-fine-tuned ^{1}H/^{13}C → SMILES generator with Claude's candidates lifts recall 33.5% → 54.1% and top-1 28.4% → 35.1% (McNemar p=0.015; Fig. S5). Zero-shot without IRexp fine-tuning recovers 0/248; with it, 25%. NMR-Solver (52.9%)[29] and trained IR transformers (63.8%)[12] confirm the task is not bound at 28–30%.

6. The decomposition across the published literature

The split measured above applies to any system reporting top-1 = a and top-k = b: recall $\geq b$, conditional precision $\leq a/b$. Table 10 applies this to every system we could read in full (`scripts/literature_decomposition.py`).

Three groups changed only the data and recall carried 68% (NMR-Solver simulated → real), 70% (SpecX random → scaffold) and 83% (Espejo education → industrial) of each collapse. Published gains come mainly from candidate supply (NMRAgent ablation[30], NMRGym formula ablation[6]). Recall binds where spectra are real and heterogeneous (87–91% of our loss); ranking dominates on simulated or single-library data (38–39%). The field almost never reports whether the true structure was proposed at all ([7] excepted).

7. Discussion

Frontier LLMs are good verifiers (89% conditional precision) and weak proposers (34% recall) on real literature spectra. The primary contribution is chemical-information infrastructure — IRexp, IRSpectra-Bench, frozen predictions and a decomposable scorer — with a diagnosis and a bounded, training-free improvement attached; not a solved elucidator (§8).

That split held across four Claude models, a battery-electrolyte subset, four verifiers and four vendor families (§6); concurrent systems that move the same levers are surveyed in §1.1.

Reporting on IRSpectra-Bench. External work should deposit ranked SMILES under the released protocol (`docs/LEADERBOARD.md`; `scripts/score_submission.py`) and report, at minimum: (i) top-1 constitution (InChIKey connectivity), (ii) generation recall (true structure in the candidate pool before re-ranking), and (iii) verification precision conditional on recall, with bootstrap CIs and candidate budget. Top-1 alone conflates the two stages this paper separates. Nonsignificant ladder gains (forward-verification vs self-ranking, McNemar p=0.55; generate-wide top-1, p=0.34) should be cited as diagnostic, not as accuracy advances.

Table 9. Verifier comparison, conditional on recall.

verifier	60-compound arm (n=19)	full benchmark (n=65)	held-out ^{13}C MAE
solver self-ranking	14/19 (74%)	55/65 (85%)	—
deterministic HOSE *lookup*	14/19 (74%)	55/65 (85%)	3.23 ppm
learned GNN (same data)	16/19 (84%)	**59/65 (91%)**	1.70 ppm
LLM forward-verifier (§5.2)	16/19 (84%)	58/65 (89%)	—

Table 10. Recall and ranking loss from published top-*k* figures. Rows are comparable only within a correctness criterion; stereochemistry-strict and connectivity figures differ by 21 points on NMR-Solver's identical predictions. Candidate budgets are given because recall over three candidates is mechanically smaller than over ten.

system	data (*n*)	top-1	top-*k* (*k*)	recall	prec.
scored on connectivity					
this work, solver alone	literature (194)	28.4%	33.5% (3)	33.5%	84.6%
this work, + forward-verification	literature (194)	29.9%	33.5% (3)	33.5%	**89.2%**
NMR-Solver[29]	literature (450)	52.9%	67.3% (10)	≥67.3%	≤78.6%
NMRAgent[30]	literature (450)	61.6%	70.0% (10)	≥70.0%	≤88.0%
scored with stereochemistry					
this work	literature (194)	21.1%	25.8% (3)	≥25.8%	≤81.8%
NMR-Solver[29]	simulated (1,000)	66.9%	89.9% (10)	≥89.9%	≤74.4%
NMR-Solver[29]	literature (450)	31.6%	53.8% (10)	≥53.8%	≤58.7%
Espejo Morales[22]	education (236)	80.9%	90.0% (5)	≥90.0%	≤89.9%
Espejo Morales[22]	industrial (34)	20.6%	29.1% (5)	≥29.1%	≤70.9%
exact match, stereo handling not stated					
Alberts, 6–13 atoms[12]	NIST IR (3,455)	63.8%	84.0% (10)	84.0%	75.9%
Alberts, 5–35 atoms[12]	NIST IR (5,024)	59.9%	78.5% (10)	78.5%	76.4%
SpecX, random split[5]	simulated (99,439)	59.0%	81.8% (10)	81.8%	72.2%
SpecX, scaffold split[5]	simulated (99,439)	29.7%	50.6% (10)	50.6%	58.7%
IR-Agent[21]	NIST IR (905)	10.3%	21.6% (10)	21.6%	47.7%
criterion not stated					
Priessner[7]	experimental (34)	20.6%	—	26.5%	77.8%

Corpus-reweighted top-1 (15.2%) is the better estimate for an arbitrary literature draw; the released 50/50 figure remains the comparable leaderboard number. IRexp re-users should keep PMC and Chemotion licence pools separable and de-leak benchmark InChIKeys before training (§2.3).

Training is not foreclosed: IRexp fine-tuning lifts recall to 54% (§5.6). What compounds is open experimental data, honest benchmarks, and inference scaffolding that rides each new model. Two practical findings: bounded contexts with tool access may help (§4.3); use forward-predicted ^{13}C agreement to re-rank, not as an abstention gauge (§5.5).

8. Limitations

Scoring is mechanical; solver runs were transcript-audited for zero ground-truth access (transcripts on request). **(i) Consumer harness.** Runs used a consumer subscription that exposes no model snapshot, temperature, seed or thinking tier; exact inference is not reproducible. Scoring is: frozen predictions, ground truth and mechanical scorers regenerate every training-free number. The instruction text wrapping Claude solver and forward-prediction batches was not captured — only per-compound payloads are released — so the verbatim prompts in the ESI are the cross-vendor harness prompts. **(ii) Pretraining contamination** is bounded by formula-only (23% → 5%) and flat recency controls (§4.6) but not excluded; verbatim spectral strings from PMC in the prompt remain a retrieval confound. **(iii) Object type, formula and scoring.** Inputs are author-transcribed band and shift lists, not raw absorbance traces or FIDs — a different object from digitised spectra and from Espejo *et al.*[22]. The molecular formula is supplied (as from HRMS); systems such as NMIRacle[10] that take no formula solve a harder prior, so absolute accuracies are not interchangeable. Headline metrics score constitution (InChIKey connectivity); with stereochemistry, top-1 is 21.1%. **(iv) Statistical honesty.** Forward-verification vs self-ranking is unresolved (McNemar p=0.55); generate-wide top-1 likewise (p=0.34). The four-model comparison at n=24 is underpowered for adjacent ranks and carries a disclosed protocol asymmetry (§4.4). The inference ladder diagnoses a bottleneck; it is not a demonstrated accuracy advance. **(v) Missing on-bench cheminformatics baselines.** Spectro, NMIRacle, Alberts IR transformers and CASE were not scored on IRSpectra-Bench, so the LLM recall wall is not cleanly separated from harness or modality choice — left to future leaderboard work under the released InChIKey scorer. **(vi) Cross-vendor scope.** Headline n=194 is Claude Opus; the four-vendor replication is on 60 compounds. Candidate budgets differ across vendors (Claude mean 2.20 vs 3.00), so recall rankings are approximate (§4.7). **(vii) Deferred and abandoned arms.** The expert-chemist audit of elucidation outputs is frozen at `data/audit/` and formally deferred — not run. Leave-one-modality-out ablation (`noIR`/`noH`/`noC`) was specified and never completed; it is abandoned for this manuscript (`docs/MODELS.md`; ESI). The extraction-recall human audit of parser coverage (§2.3) is likewise deferred. **(viii) Scope.** Battery subset uses literature electrolyte chemistry, not operando spectra. Single-sample scoring per compound (bootstrap CIs reflect compound sampling only). Organic literature bias of PMC-OA sources.

9. Methods

Mining and resolution. PMC-OA full text was fetched from `s3://pmc-oa-opendata`, parsed deterministically, and resolved with OPSIN, RDKit and SELFIES, with an optional cached PubChem fallback.

Benchmark and agents. Problems were sampled from `irexp_resolved`, stratified by RDKit ring analysis, and deduplicated across rounds by InChIKey. A compound was admitted only if its raw ^{1}H payload — multiplicities and *J* values as printed — could be re-extracted verbatim from the PMC-OA full text of its source article (`benchmark_v2.raw_1h_for`), which puts genuine coupling information in the prompt and, unavoidably, the source article's exact string (§8). The battery-electrolyte subset (§4.5) was drawn by SMARTS filters for six electrolyte functional classes (carbonate, sulfonyl/sulfonate, nitrile, sp^{3}-C–F, phosphoryl, glyme/oligoether), balanced to eight compounds per class (48 curated; 46 scored after two yielded no parseable candidate), excluding compounds used elsewhere; it was *J*-enriched and spectrally validated. Solver and forward-prediction agents were independent Claude-Opus sub-agents under the same subscription, instructed closed-book and audited for zero web/answer access. Scoring used RDKit InChIKey-connectivity match; similarity Morgan(2, 2048) Tanimoto; forward-verification a symmetric chamfer distance over ^{13}C peak sets. The core protocol trains no model and uses no paid API; the §5.6 generator and the §5.4 learned ^{13}C verifier are the only trained components, reported as complements and fenced from the headline results.

Models and versions. Headline runs used Claude Opus via consumer subscription (2026-06-09–11); formula-only control 2026-07-28; forward-prediction extension 2026-08-07. No model snapshot or decoding parameters can be reported — the harness exposes neither (§9). Fixed are frozen per-compound outputs and mechanical scorers (`docs/MODELS.md`).

Reproducibility. Every round is frozen: questions, ground-truth answers, per-agent raw outputs, predictions and scorer outputs are released; the sampler, scorer and forward-verification harness are scripted end-to-end.

Supporting Information

Supplementary figures and extended methods appear in the Electronic Supplementary Information (`docs/paper_esi.pdf`).

Author contributions

I.Y.: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing — original draft.

R.S.: methodology, software, formal analysis, investigation, validation (trained-generator and learned-verifier probes, §5.4 and §5.6), writing — review and editing.

R.A.V.-H.: conceptualization, methodology, supervision, writing — review and editing.

Conflicts of interest

There are no conflicts to declare.

Data availability

Data, frozen predictions, scoring scripts and figure regeneration are in the project repository (github.com/IlkhamFY/spectro-agent); Table 11 lists each component. A frozen archival snapshot will be deposited on Zenodo ([TODO: 10.5281/zenodo.XXXXXXX]; DOI at proof); GitHub remains the development mirror. IRexp is also mirrored at huggingface.co/datasets/ilkhamfy/IRexp (use `data/train_no_bench.jsonl.gz` for fine-tuning without benchmark leakage). Leaderboard submissions follow `docs/LEADERBOARD.md`.

IRexp redistributes extracted numeric data only (band lists, shifts, structures, source DOIs) from the PMC Open Access Subset and Chemotion, with per-record `license` / `license_pool` stamps (commercial CC-BY/CC0 primary pool 87,617; NC* held aside; Chemotion CC-BY-SA-4.0; empty/unknown excluded from commercial Zenodo — `scripts/join_pmc_licences.py`, `scripts/split_license_pools.py`). Code is MIT. Cite the Zenodo deposit and attribute sources via each record's `source_doi`. De-leak with `contrib/generator_probe/build_exp_manifest.py` before training against the benchmark; withhold `data/audit/key.jsonl` and `data/modality/key.json` from blinded reviewers.

Table 11. Headline released artefacts. Script-level inventory lives in the repository README.

component	data and code
IRexp / `irexp_resolved`	`data/irexp/`, `data/irexp_resolved/`
benchmark + within-compound control	`data/benchmark*/` — `scripts/benchmark_v2.py`
headline scoring (Table 2)	`scripts/score_main.py`
forward-verification (+ full-bench extension)	`data/fverify/`, `data/fverify_main/` — `scripts/forward_verify.py`, `scripts/forward_verify_main.py`
generate-wide (§5.3)	`data/gw/`, `data/fverify2/`, `data/fverify_gw/` — `scripts/score_generate_wide.py`
cross-vendor (§4.7)	`data/cross_vendor/` — `scripts/cross_vendor_budget.py`, `scripts/cross_vendor_gap.py`
contamination / recency (§4.6)	`data/modality/`, `data/audit/recency_control.json` — `scripts/modality_ablation.py`, `scripts/contamination_recency.py`
figures	`docs/figures/` — `scripts/make_all_figures.sh`
trained generator + GNN verifier	`contrib/generator_probe/`, `data/fverify_gen/` — `scripts/gnn_predict.py`, `data/nmrshiftdb/gnn_c13.pt`
integrity gates	`scripts/check_manuscript.py`, `scripts/check_layout.py`

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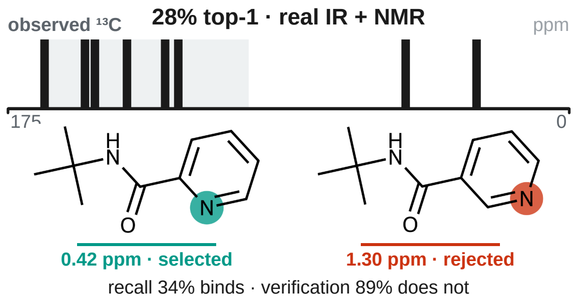

IRexp and IRSpectra-Bench: redistributable experimental IR band lists and a blind peak-list protocol — candidate recall (34%), not verification (89% once proposed), binds LLM elucidation; top-1 28% → 30% on 194 compounds.